



**UNIVERSITI TEKNOLOGI MALAYSIA
FACULTY OF COMPUTING**

TEST 2 (PRACTICAL)

SEMESTER I 2019/2020

SUBJECT CODE : SCSJ2013
SUBJECT NAME : DATA STRUCTURES AND ALGORITHMS
YEAR/COURSE : 2 (SCSJ/ SCSV/ SCSB/ SCSR/ SCSP)
TIME : 8.00 – 10.00 PM
DATE : 3rd DECEMBER 2019 (TUESDAY)
VENUE : N28 MPK1-MPK10

INSTRUCTIONS TO THE STUDENTS:

- This test consists of **TWO** questions. Answer all the questions.
- References to any resources by any means except **Data Structures and Algorithms Lab Module** are strictly prohibited.
- You are given **TWO HOURS** to complete the test inclusive the submission of your programs.
- **SWITCH OFF YOUR PHONE AND WEB BROWSER DURING THE EXAM.**

MATERIAL FOR THE TEST:

- You are provided a compressed file named **Test2(031219)_Resource.zip** containing two source code files, **Test2(031219)_Q1.cpp** and **Test2(031219)_Q2.cpp**, and the question, **Test2(031219)_Question.pdf**.
- Download the file **Test2(031219)_Resource.zip** from the UTM's elearning and decompress it into the local hard drive of your computer.
- The provided program files should be used as the basis to answer this test.

SUBMISSION PROCEDURE:

- Only the source code files (i.e. the **file with the extension .cpp**) is required for the submission.
- Submission must be done via the **UTM's e-learning system**.

Question 1

[30 Marks]

Given a C++ program file, **Test2(031219)_Q1.cpp**, which creates a linked list with the initial state as shown in Figure 1. The variables **head** and **last** are pointers that pointing to the first and last node of the list, respectively.

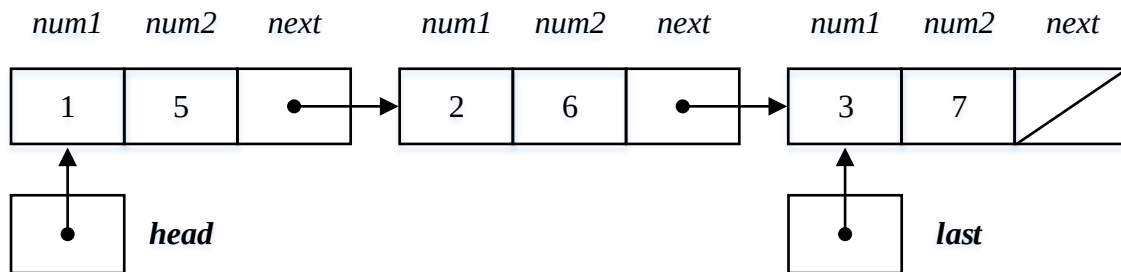


Figure 1: The initial state of the list

Append the appropriate lines of code at the space provided in **Test2(031219)_Q1.cpp** to accomplish each of the following tasks. Note that, each task is continuous from one to the next. You must not change any line of code regarding the class definition and the given lines of code in the **display** and **main** function. Figure 7 shows the expected output when the program runs.

Task 1: Write the code to calculate and display the **sum** of *num1* value in the first and second node. (4 marks)

Task 2: Write the code to add 3 to *num2* value in all the nodes. Figure 2 shows the result of this task. **Note:** In this task, you should use **loop**. (6 marks)

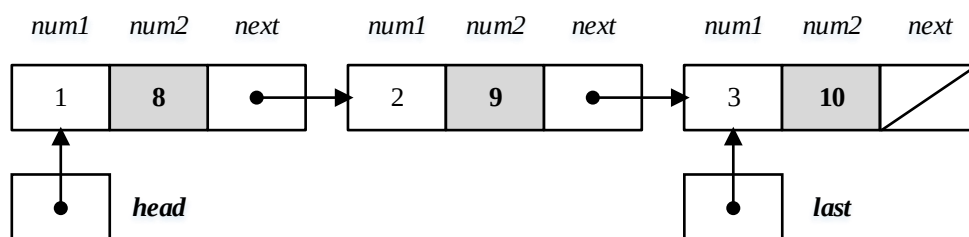


Figure 2

Task 3: Write the code to add new node with data *num1* = 5 and *num2* = 12 and set it as the new *head*. The result is given in Figure 3. (4 marks)

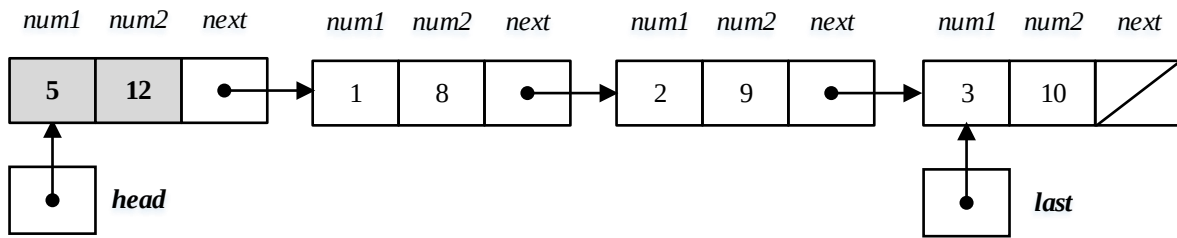


Figure 3

Task 4: Write the code to set the head as the last node and the second node as the new head node.

The result is given in Figure 4.

(4 marks)

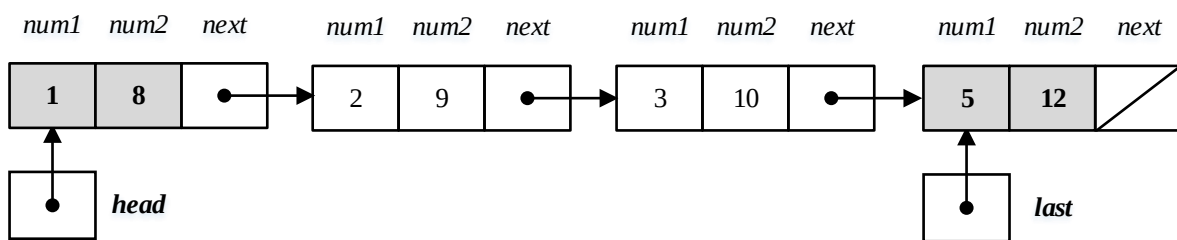


Figure 4

Task 5: Write the code to insert new node with data **num1 = 4** and **num2 = 11** before the last node. Figure 5 shows the result of this task. **Note:** In this task, you should use **loop**.

(8 marks)

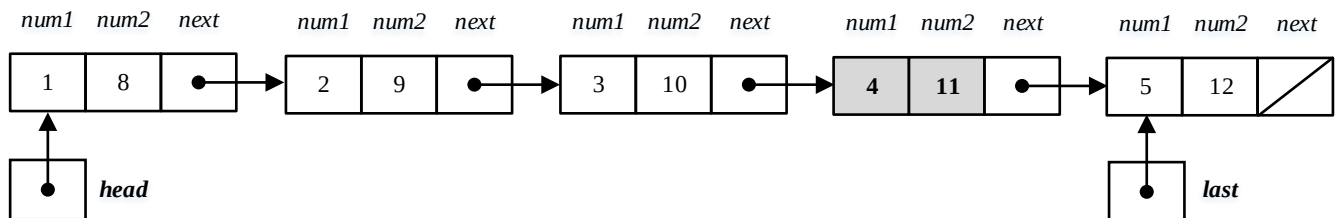


Figure 5

Task 6: Write the code to delete the second node in the list. Figure 6 shows the result of this task.

(4 marks)

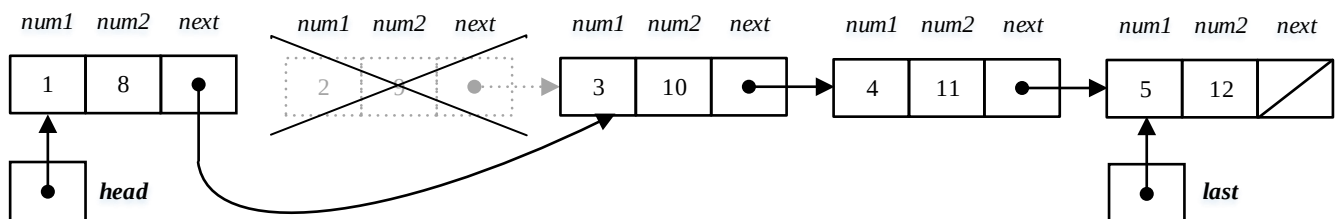


Figure 6

```
Traverse and display linklist data
|1 5|    |2 6|    |3 7|

Executing Task 1 - calculate the sum of num1 value for the first and
second node

Sum of num1 value in first and second node is equal to: 3

Executing Task 2 - add 3 to num2 value in all the nodes

Traverse and display linklist data
|1 8|    |2 9|    |3 10|

Executing Task 3 - add new node(5, 12) as new head

Traverse and display linklist data
|5 12|   |1 8|    |2 9|    |3 10|

Executing Task 4 - set head to last and set second node as new head

Traverse and display linklist data
|1 8|    |2 9|    |3 10|   |5 12|

Executing Task 5 - insert new node(4, 11) before last node

Traverse and display linklist data
|1 8|    |2 9|    |3 10|   |4 11|   |5 12|

Executing Task 6 - delete the second node

Traverse and display linklist data
|1 8|    |3 10|   |4 11|   |5 12|
```

Figure 7: The output of the program for Question 1

Question 2

[30 Marks]

Given a C++ program file, **Test2(031219)_Q2.cpp**, which contains a linked list implementation of stack to manage a car(s) in Munir Garage. The garage allow ten cars to be parked in a space that would usually fit only one. The definition of the class **nodeStack** and **stack** are given in the program file. The variable **top** in the class **stack** is a pointer which points to the top node of the stack. The implementation of the method **createStack()** and **isEmpty()** of the class **stack** are also given in the **Test2(031219)_Q2.cpp**. You must not change any line of code regarding the definition of the class **nodeStack** and **stack**.

Complete the implementation of **push()**, **pop()**, **display()** and **search()** function by appending the appropriate lines of code at the space provided in **Test2(031219)_Q2.cpp** to accomplish each of the following tasks:

Task 1: Write the code to insert new car to the stack. Firstly, ask the user to enter the information of the car which is the plate number, brand/ model and owner's name. **Note:** Please use the variables defined in line 59. Then, insert the new car by using the information entered (please use the object pointer defined in line 60). **(8 marks)**

Task 2: Write the code to remove the car from the stack (please use the object pointer defined in line 94). **(3 marks)**

Task 3: Write the code to display all the cars in the stack (please use the object pointer defined in line 121). **(9 marks)**

Task 4: Write the code to search the car in the stack based on brand/ model (please use the object pointer defined in line 151). **(10 marks)**

Figure 8(a) and 8(b) show the expected output when the program runs. **Note:** The values in **bold** are input by the user.

```

MUNIR GARAGE MANAGEMENT SYSTEM

1. Insert car
2. Remove car
3. Search car
4. Display cars
5. Exit

Enter your choice [1-5]: 1

<<<<<< Insert Car >>>>>>

Plate Number : JPQ1234
Brand/ Model : Nissan
Owner       : Sofea Ali

MUNIR GARAGE MANAGEMENT SYSTEM

1. Insert car
2. Remove car
3. Search car
4. Display cars
5. Exit

Enter your choice [1-5]: 1

<<<<<< Insert Car >>>>>>

Plate Number : MAX4567
Brand/ Model : Honda
Owner       : Anis Ali

MUNIR GARAGE MANAGEMENT SYSTEM

1. Insert car
2. Remove car
3. Search car
4. Display cars
5. Exit

Enter your choice [1-5]: 1

<<<<<< Insert Car >>>>>>

Plate Number : VAT8976
Brand/ Model : Nissan
Owner       : Arif Ali

```

```

MUNIR GARAGE MANAGEMENT SYSTEM

1. Insert car
2. Remove car
3. Search car
4. Display cars
5. Exit

Enter your choice [1-5]: 4

<<<<<< Display Cars >>>>>>

[1] Plate Number : VAT8976
    Brand/ Model : Nissan
    Owner       : Arif Ali

[2] Plate Number : MAX4567
    Brand/ Model : Honda
    Owner       : Anis Ali

[3] Plate Number : JPQ1234
    Brand/ Model : Nissan
    Owner       : Sofea Ali

<< Press any key to continue >>

MUNIR GARAGE MANAGEMENT SYSTEM

1. Insert car
2. Remove car
3. Search car
4. Display cars
5. Exit

Enter your choice [1-5]: 3

<<<<<< Search Car(s) >>>>>>

Enter brand/ model: Nissan

Plate Number : VAT8976
Brand/ Model : Nissan
Owner       : Arif Ali

Plate Number : JPQ1234
Brand/ Model : Nissan
Owner       : Sofea Ali

<< Finish searching... Press any key
to continue >>

```

Figure 8(a): The output of the program for Question 2

```
MUNIR GARAGE MANAGEMENT SYSTEM

1. Insert Car
2. Remove car
3. Search car
4. Display cars
5. Exit

Enter your choice [1-5]: 2

<<<<<<< Remove Car >>>>>>>

Plate Number : VAT8976
Brand/ Model : Nissan
Owner       : Arif Ali

<< Press any key to continue >>
```

```
MUNIR GARAGE MANAGEMENT SYSTEM

1. Insert car
2. Remove Car
3. Search car
4. Display cars
5. Exit

Enter your choice [1-5]: 4

<<<<<<< Display Cars >>>>>>>

[1] Plate Number : MAX4567
    Brand/ Model : Honda
    Owner       : Anis Ali

[2] Plate Number : JPQ1234
    Brand/ Model : Nissan
    Owner       : Sofea Ali

<< Press any key to continue >>
```

Figure 8(b): The output of the program for Question 2